

PHO / RICNAC Dynamics CRM Solution Architecture

Old Solution Structure Guide for Migration Planning

Prepared from the local Codes workspace | May 20, 2026

Document scope

This guide documents the legacy CRM solution architecture only: core folders, plugin assemblies, JavaScript web resources, reporting/migration assets, and upgrade notes relevant to migration analysis. It does not perform deep code walkthroughs or define final target-state migration mappings.

1. Executive Summary

What this solution is: A legacy Microsoft Dynamics CRM enterprise customization workspace for PHO / RICNAC. It contains server-side plugin DLLs, CRM form JavaScript web resources, SSRS report definitions, shared CRM helper libraries, test code, and supporting migration/configuration assets.

Business/system purpose: The solution appears to support CRM operations around contacts, organizations/accounts, education, incidents/inquiries, campaigns, attendance, service requests, reporting, and historical migration data handling.

Main technology stack: Microsoft Dynamics CRM 2013-era SDK assemblies, Microsoft.Xrm.Sdk, C# .NET Framework class libraries, JavaScript Xrm form scripting, SSRS .rdl reports, SQL/Access/CSV/XML/Excel migration artifacts, Visual Studio/MSBuild, and the CRM Plugin Registration Tool.

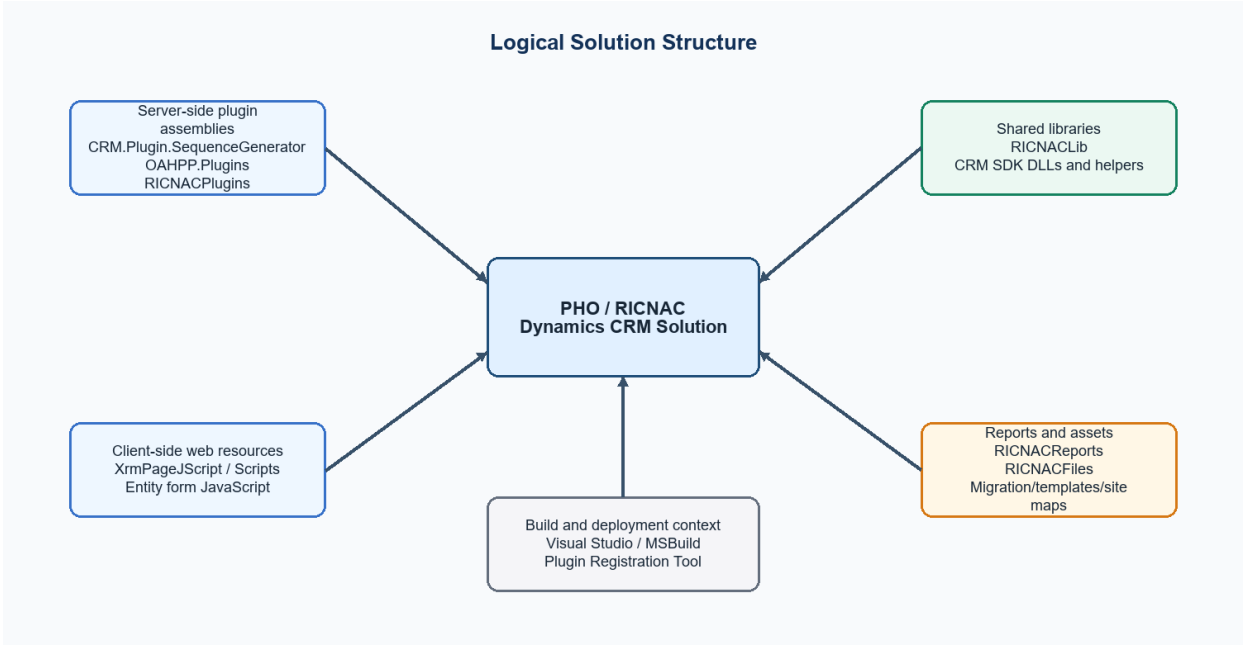
High-level risks: The codebase depends on legacy CRM SDK assemblies and older CRM patterns. It includes .NET Framework plugin projects, SSRS report projects that are not buildable by plain MSBuild, and large migration assets that may include historical or sensitive data. Plugin registration metadata is not fully represented in source control, so runtime behavior must be confirmed against the source CRM environment.

Area	Primary Purpose	Migration Insight
Plugin assemblies	Server-side business rules executed by CRM pipeline.	Inventory and registration steps are required before migration.
JavaScript web resources	Client-side form behavior for CRM records.	May require modernization from legacy Xrm.Page patterns.
Reports	SSRS .rdl reporting over CRM data.	Report data sources and queries must be reviewed for Dataverse compatibility.
Migration assets	CSV/XML/Excel/Access/ZIP files used for historical data movement.	Must be classified, cleansed, and mapped before target migration.

2. Architecture Overview

The old solution is arranged around Dynamics CRM customizations. CRM records sit at the center; client scripts shape form behavior, server-side plugins enforce backend logic, reports consume CRM data, and migration/configuration assets support historical import/export or deployment tasks.

High-Level CRM Flow Diagram



Server-Side Plugin Flow

- A CRM event such as create, update, assign, or send email is registered as a plugin step.
- The CRM runtime invokes the plugin assembly and passes `IServiceProvider` into `Execute(...)`.
- The plugin reads `IPluginExecutionContext`, `ITracingService`, and `IOrganizationService`.
- The plugin validates the Target entity, performs business logic, and uses `Microsoft.Xrm.Sdk` services to read or update records.
- Failures are surfaced through `InvalidPluginExecutionException` or CRM tracing, depending on the implementation.

Client-Side JavaScript Flow

- A CRM user opens or edits a record form.
- Entity-specific JavaScript web resources run on form load, field change, or save events.
- Shared libraries provide common helpers and CRM service access patterns.
- The scripts manipulate fields, validations, visibility, lookups, or related data using the legacy Xrm client API.
- When the record is saved, server-side plugins may also execute in the CRM pipeline.

Reporting and Data Migration Flow

- `RICNACReports` contains SSRS `.rdl` files for operational reporting.
- `RICNACFiles` contains data migration folders, Access databases, XML/CSV/Excel files, templates, site maps, and ZIP archives.
- Reports and migration files depend on entity schema, field names, relationships, and historical data conventions.
- For migration planning, these assets should be treated as separate from executable plugin code but still part of the enterprise solution footprint.

How Components Relate

Component	Relationship to CRM Records	Migration Consideration
Plugins	Run when records are created, updated, or processed.	Confirm target event registrations and whether plugins should be disabled during bulk migration.

JavaScript web resources	Run on CRM forms before users save data.	Review for deprecated Xrm.Page usage and form event compatibility.
Reports	Read CRM data for operational output.	Validate data source, query, and security changes in Dataverse.
Migration assets	Represent historical/import data and supporting templates.	Classify source data, map entities/fields, and confirm whether assets are still active.

3. Plugin Documentation

Plugin registration step metadata is not fully present in the repository. Where trigger entity is listed, it is inferred from source comments, class documentation, or file naming. CRM message and stage should be verified from the source CRM Plugin Registration Tool export or environment metadata.

Assembly	Main Classes / Scope	Dependencies	Deployment Notes
CRM.Plugin.SequenceGenerator	SequenceGenerator, SequenceHelper. Generates configured sequence/autonumber values for CRM records.	Microsoft.Xrm.Sdk	Register per target entity/message. Requires ce_counter configuration records. The project now uses a relative HintPath to the local SDK DLL.
OAHP.Plugins	Attendance, campaign email/list automation, and service request plugins.	Microsoft.Xrm.Sdk, Microsoft.Crm.Sdk.Proxy	Registration steps are not stored in source. Project now uses CrmSdkLibDir to resolve local CRM SDK DLLs.
Digitalembbrace.RICNACPlugins	Main PHO/RICNAC plugin assembly covering account, contact, education, incident, document location, sharing, validation, and helper classes.	Microsoft.Xrm.Sdk, Microsoft.Crm.Sdk.Proxy, Microsoft.Xrm.Sdk.Workflow, service references	Signed assembly. Registration/deployment must be confirmed from CRM environment or deployment records.

Known Plugin Classes

Assembly / File	Trigger Entity	Purpose	Message / Stage Notes
CRM.Plugin.SequenceGenerator / SequenceGenerator.cs	Dynamic, based on ce_counter ce_name	Generates a next sequence value and writes it to the configured target attribute.	Message/stage not in repo. Code checks IsInTransaction and should be registered where target attributes can be set safely.
CRM.Plugin.SequenceGenerator / SequenceHelper.cs	ce_counter	Reads sequence configuration and persists incremented counter values.	Helper class, not directly registered as a plugin.
OAHP / AttendanceUpdateSumUpNumAttendances.cs	new_attendance	Recalculates campaign attendee count based on matching attendance records.	Message/stage not in repo; likely triggered by attendance create/update.
OAHP / CampaignCreateAttendances.cs	campaign	Creates attendance records for contacts in campaign-related marketing lists.	Message/stage not in repo; verify whether create or update is intended.
OAHP / CampaignActivitySendAllEmails.cs	campaignactivity	Sends pending email activities related to a campaign activity.	Uses SendEmailRequest; registration must be confirmed.
OAHP / SRRequestCreateSetName.cs	new_servicerequestform	Builds new_name from first and last name.	Likely pre-operation if target field assignment should persist automatically.
OAHP / SRResponseCreateSearchContact.cs	new_servicerequestresponse	Finds contact by email and copies selected contact values into the response.	Message/stage not in repo.
OAHP / SRFormUpdateCreateSRResponse.cs	new_servicerequestform	Creates a service request response record from service request form data.	Message/stage not in repo.
RICNACPlugins / Account.cs	account	Account validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / AssignAutoNumber.cs	various/custom	Autonumber assignment behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / BatchUpdateActivity.cs	activity	Batch activity update behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.

RICNACPlugins / CampaignResponse.cs	campaignresponse	Campaign response validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / CommunicationListActivity.cs	communication/list activity	Communication list activity behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / Contact.cs	contact	Contact validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / DeleteActivity.cs	activity	Activity delete-related behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / DocumentLocation.cs	document location	Document location validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / Education.cs	education	Education record validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / EducationAttendee.cs	education attendee	Education attendee validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / Incident.cs	incident/case	Incident or inquiry validation/processing behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / ManageMemberActivity.cs	member/activity	Membership or activity management behavior.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / ProcessEntity.cs	shared	Shared entity processing support.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / SetEntityName.cs	various	Sets display names for configured entities.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / ShareActivity.cs	activity	Shares activity records.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / UnShareActivity.cs	activity	Removes activity sharing.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / ValidateEntity.cs	shared	Shared validation support.	Registration/message/stage not stored in repo; verify from CRM plugin registration.
RICNACPlugins / BaseEntity.cs / PluginHelper.cs / XrmBasic.cs	shared	Base classes, helpers, and CRM proxy/support code.	Registration/message/stage not stored in repo; verify from CRM plugin registration.

Plugin migration insight

Before migrating or re-registering plugins, export the source CRM plugin registration steps. The source code alone does not prove the message, filtering attributes, stage, execution order, impersonating user, secure/unsecure config, or isolation mode.

4. JavaScript Web Resources

The XrmPageJScript project contains JavaScript web resources intended for Dynamics CRM forms. Entity-specific files appear to map to CRM record forms, while shared libraries provide reusable helpers or legacy service access wrappers.

JavaScript File	Likely CRM Entity/Form	Type	Purpose / Notes
de_Account.js	Account	Entity-specific	Account form behavior, validation, and field logic.
de_Common.Library.js	Multiple forms	Shared library	Common reusable helper functions for CRM scripts.
de_CommunicationList.js	Communication List	Entity-specific	Communication list form behavior.
de_Contact.js	Contact	Entity-specific	Contact form behavior and validation.
de_Education.js	Education	Entity-specific	Education form behavior.
de_EventAttendee.js	Event Attendee	Entity-specific	Event attendee form behavior.
de_Incident.js	Incident / Case	Entity-specific	Incident/inquiry form behavior.
de_Incident_Quick.js	Incident quick form	Entity-specific	Likely quick-create or compact incident form behavior.
new_incident_scripts.js	Incident / Case	Entity-specific	Additional incident scripting, possibly legacy customizations.
oahpp_commonJsript.js	Multiple forms	Shared library	OAHP-specific common form utilities.
XrmPageTemplate.js	Template	Template/support	Template for Xrm page scripting patterns.
XrmServiceToolkit.js	Multiple forms	Toolkit/vendor-style	Legacy CRM service access helper; review compatibility during modernization.
XrmSvcToolkit.js	Multiple forms	Toolkit/vendor-style	Legacy CRM service helper; review overlap with XrmServiceToolkit.

Xrm Client API Notes

- This is a legacy CRM form scripting area. Older files may use Xrm.Page or SOAP/OData helper patterns that require review for modern model-driven apps.
- During migration, confirm each JavaScript web resource is still referenced by forms, command bars, ribbons, or event handlers.
- Shared libraries should be loaded before entity-specific scripts when forms depend on common helper functions.
- For Dynamics 365 CE / Dataverse modernization, validate formContext usage, async API behavior, security role assumptions, and deprecated client methods.

5. Upgrade Notes

Projects Upgraded to .NET Framework 4.7.1

Project	Previous Target	Current Target	Verification
CRM.Plugin.SequenceGenerator	v4.0 / invalid v7.1 during review	v4.7.1	Build succeeded.
Digitalembbrace.RICNACLib	v4.0	v4.7.1	Build succeeded with pre-existing warnings.
Digitalembbrace.RICNACPlugins	v4.0	v4.7.1	Build succeeded with pre-existing warnings.
Digitalembbrace.RICNACTest	v4.0	v4.7.1	Build succeeded with pre-existing warnings.

XrmPageJScript	v4.5.1	v4.7.1	Build succeeded.
OAHP.Plugins	v4.0	v4.7.1	Build succeeded after CRM SDK reference path fix.

Why .NET 7 / v7.1 Was Not Correct

- TargetFrameworkVersion values such as v4.7.1 refer to .NET Framework targets. v7.1 is not a valid .NET Framework target.
- .NET 7 uses SDK-style project files and TargetFramework values such as net7.0, which is a different runtime model.
- Classic Dynamics CRM plugin assemblies are .NET Framework class libraries, not standalone .NET 7 applications.
- The old CRM SDK dependency set in this repository is aligned to legacy Microsoft.Xrm.Sdk assemblies, not modern .NET 7 application deployment.

CRM Compatibility Warnings

- Local build success does not guarantee deployment compatibility with every CRM server version.
- The target CRM/Dynamics 365 CE environment must support plugin assemblies compiled for .NET Framework 4.7.1.
- The bundled Microsoft.Xrm.Sdk.dll is version 6.0.0.0, which indicates an older CRM SDK generation.
- Plugin behavior may depend on registration metadata that is not included in the source files.

Reference Path Changes

- CRM.Plugin.SequenceGenerator now resolves Microsoft.Xrm.Sdk.dll from DigitalEmbrace_CRM_Solutions/RICNACLib/lib using a relative HintPath.
- OAHP.Plugins now uses a CrmSdkLibDir MSBuild property to locate Microsoft.Crm.Sdk.Proxy.dll and Microsoft.Xrm.Sdk.dll.
- RICNACLib, RICNACPlugins, and RICNACTest already referenced local SDK DLLs through relative HintPath entries.

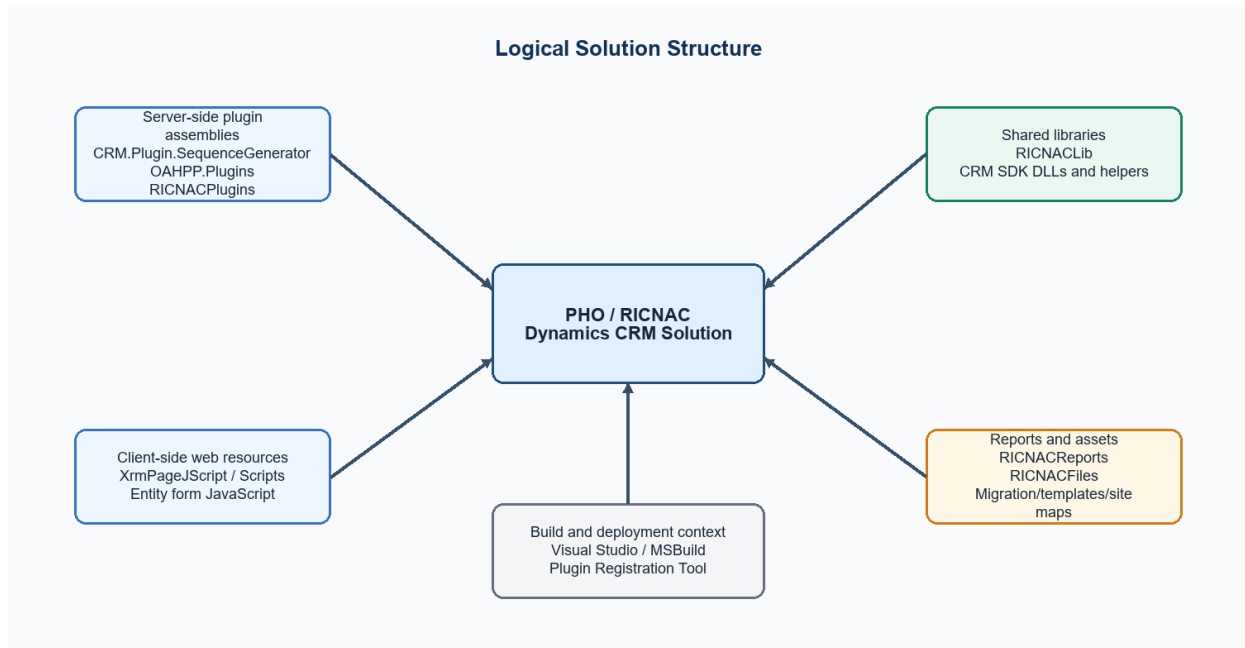
Remaining Technical Debt

- Plugin registration exports are not present in the repository and should be collected before migration.
- The full PHO solution does not build cleanly with plain MSBuild because of legacy RICNACReports.rptproj and RICNACFiles metaproject behavior.
- Generated service references, old SDK DLLs, and legacy JavaScript toolkit files should be reviewed before target-state migration.
- Migration assets may contain historical or sensitive data and should be classified before being moved to a new environment or repository.
- Existing code warnings include unused variables and hidden inherited methods; these were not addressed as part of the framework upgrade.

6. High-Level Diagrams

Logical Folder and Solution Flow

A good first diagram for this project is a high-level CRM solution flow showing how the main folders relate logically. It separates server-side plugins, client-side scripts, shared libraries, reports, and migration/configuration assets.



Runtime-Oriented Flow

A more runtime-oriented flow shows how a CRM user action moves through form scripts, CRM record operations, plugin execution, SDK calls, and reporting or migration-related data outputs.

